

Reliability-Aware Gated Distillation from Multi-Branch Teachers for Multimodal Rumor Detection

Anonymous ACL submission

Abstract

Detecting multimodal misinformation requires combining textual and visual cues while coping with topic shift and annotation noise. We present Dynamic Modal Priority Distillation (DMPD), a reliability-aware gated distillation framework that routes supervision from a multimodal teacher to a student only when (i) a fused prediction is confident and self-consistent across modalities, and (ii) the post belongs to an event with a good reliability history. The objective emphasizes calibration and positive-class recall via temperature-scaled KL, lightweight feature alignment, and evidential regularization. Unlike standard distillation that copies teacher predictions indiscriminately, our design selectively transfers knowledge across modalities and events, reducing the propagation of confident errors. We evaluate on two Chinese multimodal rumor datasets (Weibo and WeFEND) and a synthetic English benchmark (MiRAGE-News) with strong single- and multimodal baselines, and provide ablations on gating, event reliability, and calibration. Code, configurations, and scripts are released for full reproducibility.

1 Introduction

Misinformation spreads through short, image-rich posts where cues in text and vision interact in subtle ways. Multimodal detectors improve over text-only systems but remain vulnerable to topic shift and annotation noise; the literature explores event-invariant representations (Wang et al., 2018), similarity-aware fusion (Zhou et al., 2020), graph/social context (Lu and Li, 2020; Nguyen et al., 2020), and large-scale benchmarks such as Fakeddit (Nakamura et al., 2020). In parallel, the vision-language community has shifted toward pretrained encoders (e.g., ViLBERT, CLIP)

that better capture cross-modal semantics (Lu et al., 2019; Radford et al., 2021), yet deploying them robustly on domain-specific misinformation remains challenging.

Knowledge distillation (KD) transfers supervision from accurate or expensive teachers to a compact student (Hinton et al., 2015; Gou et al., 2021), including mutual/self-distillation variants (Zhang et al., 2018; Furlanello et al., 2018). However, in noisy, shifting domains like misinformation detection, teachers can be wrong with high confidence; naively imitating them propagates errors and degrades calibration. Robust learning under noise selects reliable instances or mentors during training (Han et al., 2018; Jiang et al., 2018), but comparable *selective* mechanisms for distillation from *multimodal teachers* are under-explored.

Uncertainty and calibration are crucial for safety and downstream moderation. Modern neural networks are often miscalibrated (Guo et al., 2017); evidential learning and prior networks explicitly model predictive uncertainty (Sensoy et al., 2018; Malinin and Gales, 2018). For KD, temperature control and cost-sensitive objectives can mitigate over-confident transfer, especially on rare positives that matter most to users.

In this work we focus on a concrete and practically important setting: supervised multimodal rumor detection with *multi-branch teachers and explicit event identifiers*. We assume a fixed multimodal teacher with text, vision, and fusion branches (as in EANN/SAFE-style encoders) and event IDs for each post (rumor/story IDs in Weibo; public-account stories in WeFEND). Our goal is not a universal KD framework, but to study how selective, reliability-aware transfer from such teachers can reduce error propagation under event-structured noise.

We propose **Dynamic Modal Priority Distillation (DMPD)**, a reliability-aware *gated multimodal distillation* scheme for multimodal misinformation detection. DMPD routes supervision from a multimodal teacher with text, vision, and fusion branches to the student only when (i) a *teacher-match gate* confirms agreement and confidence, and (ii) an *event-level reliability* score indicates consistent teacher correctness. A distillation objective combines temperature-scaled KL, lightweight feature alignment, and positive-class costs with evidential heads. Compared to prior multimodal detectors that focus on new backbones or fusion, DMPD focuses on *how* to transfer supervision from multiple imperfect views of a single multimodal teacher in this event-aware setting. Compared to classical KD that treats teacher outputs as fixed soft labels, DMPD uses agreement- and reliability-aware *selection* and *calibration* to reduce error propagation.

Our contributions are:

- **Reliability-aware gated distillation:** a lightweight, agreement-based gate that routes knowledge only when the text, vision, and fusion branches of a multimodal teacher agree and are confident, paired with an event reliability memory that suppresses bursty, error-prone topics in supervised rumor datasets.
- **Event-level stability filtering:** event-level reliability estimates for Weibo and WeChat/WeFEND based on explicit story/rumor IDs, and a simple filtering scheme that down-weights historically unreliable events during training.
- **Calibration-aware objective:** a temperature-annealed KD loss with evidential heads and positive-class costs that improves macro-F1 while avoiding degradation of Expected Calibration Error (ECE), and on WeFEND yields modest ECE reductions.
- **Empirical validation in a specific setting:** experiments on two Chinese multimodal datasets (Weibo, WeChat/WeFEND) with strong baselines (EANN, SAFE, fusion teachers), three- to five-seed statistics where available, and

ablations on gating/reliability schedules, plus preliminary extensions to an English synthetic benchmark (MiRAGeNews) summarized in Appendix B. We also report strict time-split Fakeddit results as a hard case where weak teachers limit the benefits of our scheme.

Comprehensive experiments, ablations, and visualizations are detailed in Sections 5–6. We focus on settings where a reasonably strong multimodal teacher is available and aim to make distillation safer and better calibrated in the presence of event-level noise, rather than rescuing arbitrarily weak teachers. Code, configs, and scripts will be released for full reproducibility once anonymity is lifted.

2 Related Work

Multimodal Misinformation Detection.

Event-invariant representation learning improves topic robustness (EANN; Wang et al., 2018). Subsequent work enhanced fusion and context modeling via similarity-aware architectures (Zhou et al., 2020) and graph/social signals for explainability and early detection (Lu and Li, 2020; Nguyen et al., 2020). Data resources evolved from platform-specific rumor corpora to larger multimodal benchmarks such as Fakeddit (Nakamura et al., 2020) and FakeNewsNet (Shu et al., 2020), enabling finer-grained evaluation. In parallel, vision-language pretraining (e.g., ALBEF/BLIP/BLIP-2) (Li et al., 2021, 2022, 2023) and CLIP-style models (Radford et al., 2021) improved cross-modal alignment and provide strong encoders for downstream detection. Our focus is orthogonal: rather than new backbones or graph modules, we study *selective transfer* from imperfect teachers in a reliability-aware way.

Knowledge Distillation and Self-Training.

KD compresses heavy models into compact students (Hinton et al., 2015; Gou et al., 2021); mutual/self-distillation can further enhance students (Zhang et al., 2018; Furlanello et al., 2018). However, KD may propagate teacher mistakes under distribution shift. Robust training under label noise (e.g., Co-teaching and data-driven curricula) (Han et al., 2018; Jiang et al., 2018) motivates *selecting* reliable supervision. Self-training with Noisy Student

(Xie et al., 2020) shows that pseudo-labels benefit from confidence control and strong augmentation—a perspective shared by our agreement- and reliability-gated transfer.

Uncertainty and Calibration. Modern networks are miscalibrated (Guo et al., 2017); temperature scaling is a strong post-hoc baseline. Evidential learning (Sensoy et al., 2018) and prior networks (Malinin and Gales, 2018) provide principled uncertainty estimates. Label smoothing improves generalization and sometimes calibration (Müller et al., 2019). We integrate calibration into training via temperature-aware KL and evidential regularization, and explicitly target positive-class recall, complementing post-hoc scaling.

Recent Trends (2024–2025). Large vision–language models (VLMs) and multimodal LLMs are increasingly used as strong encoders or as teachers for downstream detection, often paired with retrieval or external tools to improve factual grounding. Synthetic data and instruction-style supervision have been explored to bridge distribution shift and expose rare phenomena, while new evaluations emphasize calibration, out-of-distribution robustness, and error analysis beyond single accuracy numbers. Our work is complementary: rather than relying on ever-larger teachers, we focus on *selective, reliability-aware transfer* that improves a compact student under topic shift and noisy supervision, and we evaluate with calibration- and event-aware analyses.

3 Method

3.1 Problem Setup and Overview

We consider a single, fixed multimodal teacher with three branches: a text branch $f^{(t)}$, a vision branch $f^{(v)}$, and a lightweight fusion branch $f^{(f)}$. For notational convenience we refer to these branches as “teachers” or “views.” The student s_θ shares the same encoders with a fusion head of comparable capacity, but only the student parameters are updated during distillation. For a post $x = (x_t, x_v)$ with label $y \in \{0, 1\}$, the task is binary misinformation detection. Let $p^{(k)}(y | x)$ be the posterior of branch $k \in \{t, v, f\}$ and $p_\theta(y | x)$ that of the student. We denote the predicted class and confidence of each teacher branch by $(\hat{y}_k, c_k)$,

where c_k is a temperature-scaled softmax score used only for gating and reliability.

Two coupled phenomena motivate *selective* knowledge transfer. First, *cross-modal inconsistency*: text and image may conflict (e.g., satire text plus unrelated stock photo), so copying whichever teacher is more confident can amplify errors. Second, *event burstiness*: posts clustered in an event/topic e may share correlated noise. Our design therefore has two filters with distinct granularity: an **instance-level gate** that requires cross-view agreement, and an **event-level reliability** that down-weights unreliable topics. Throughout, the selected modality $m \in \{t, v\}$ is chosen per instance by a lightweight selector (default: the teacher with higher calibrated confidence), while the fusion teacher provides a consensus view.

3.2 Teacher-Match Gating

For brevity, denote the fused teacher by $(\hat{y}_f, c_f)$ and the selected-modality teacher by $(\hat{y}_m, c_m)$. The gate maps an instance to $\{0, 1\}$:

$$g(x) = \begin{cases} 1, & \text{if } \hat{y}_f = \hat{y}_m, c_f \geq \tau_f, \\ & c_m \geq \tau_m, |c_f - c_m| \leq \delta, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Here δ is a confidence-gap margin and temperatures T modulate softmax sharpness. Intuitively, the gate admits a sample only when (i) fusion and the selected modality predict the *same* class and (ii) both are confident without disagreeing in magnitude. This conservative rule privileges cross-view consensus and avoids copying high-confidence mistakes (Blum and Mitchell, 1998). When the gate is open ($g(x) = 1$), the distillation target is taken from the teachers; otherwise the student falls back to supervised terms only (cf. the objective below).

3.3 Event-Level Reliability

Each post is assigned to an event/topic e (rumor or news story IDs, conversation roots on threaded platforms, or public-account stories in WeChat/WeFEND). In our experiments we use the explicit event IDs provided by Weibo and WeFEND; for datasets without such IDs (e.g., MiRAGeNews) we do not apply event-level filtering and only use instance-level gating. We track a reliability score $r_e \in [0, 1]$ by

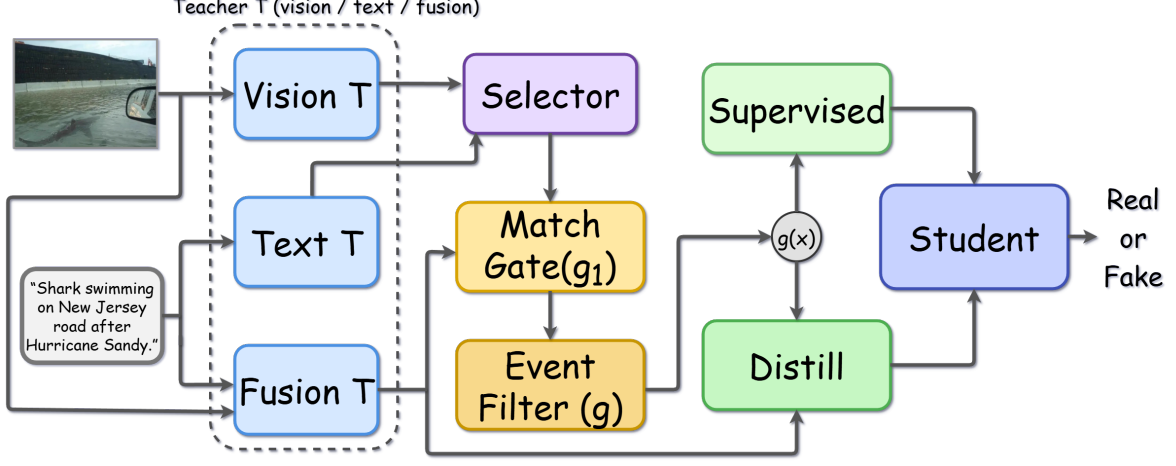


Figure 1: Overview of Dynamic Modal Priority Distillation (DMPD). The dashed box denotes the multimodal teacher with vision, text, and fusion branches. A selector picks the more reliable unimodal branch, and the match gate checks its agreement with the fusion branch. The event filter incorporates event-level reliability to produce the final gate $g(x)$, which controls whether the student receives distillation from the fusion teacher or is trained only with supervised losses. At test time, only the student is used to predict real or fake.

an exponential moving average (EMA) over past teacher outcomes within that event: $r_e \leftarrow (1 - \beta) r_e + \beta \mathbb{I}[\text{teacher correct}]$, where correctness is measured against ground-truth labels on the training set only. We also maintain a companion EMA of mean confidence. During training, if the event reliability drops below a cutoff or confidence is persistently low, we force the gate to close for that event (i.e., treat it as $g(x) = 0$ irrespective of instance cues). This decouples learning from bursts of correlated noise and echoes curriculum/mentor ideas under noise (Jiang et al., 2018; Han et al., 2018). At test time, r_e is not updated.

3.4 Calibration-Oriented Objective

Let q be the teacher target used when the gate admits a sample (fusion or the selected modality), and let $h_\theta, h^{(f)}$ be the student and fusion features. We anneal temperatures and margins over the training fraction $\lambda \in [0, 1]$. The objective below decomposes into short, named terms to make the roles of distillation, supervision, feature alignment and calibration explicit:

$$\begin{aligned} \mathcal{L} &= \lambda_{ce} L_{ce} + \lambda_{kl} L_{kl} + \lambda_{feat} L_{feat} + \lambda_+ L_+ + \lambda_{evi} L_{evi}, \\ \text{where } L_{kl} &= \text{KL}(q_T \| p_{\theta, T}), \\ L_{feat} &= \|h_\theta - h^{(f)}\|_2^2, \\ L_+ &= \mathbf{1}\{y = 1\} \text{CE}(p_\theta, y), \\ L_{ce} &= \text{CE}(p_\theta, y) \end{aligned} \quad (2)$$

where temperature T controls sharpness, the positive-class term combats imbalance, and $\mathcal{L}_{evi}$ discourages unwarranted certainty (Sensoy et al., 2018). When the gate is closed, the loss reduces to the supervised components (L_{ce} and L_+); when open, the KL and feature alignment terms engage. For reporting calibration, we apply post-hoc temperature scaling on validation as in Guo et al. (2017).

Evidential Head. We parameterize a Dirichlet distribution over class probabilities with concentration $\alpha = e + \mathbf{1}$, where nonnegative evidence e is predicted by a small head. The total evidence $S = \sum_c \alpha_c$ controls confidence. We use a standard evidential loss that combines a supervised term (encouraging large evidence for the true class) with a regularizer that penalizes excessive evidence on misfit examples, promoting calibrated uncertainty (Sensoy et al., 2018).

3.5 Optimization and Scheduling

We use a simple schedule that mirrors the roles of the gate and reliability: *warm-up* (no KD; stabilize supervised prediction), then *distillation* with progressively tighter thresholds and cooler temperatures, and finally an optional *fine-tune* (no KD) for calibration. Concretely, thresholds $\tau_f(\lambda), \tau_m(\lambda)$ increase and the margin $\delta(\lambda)$ decreases with λ ; the event cutoff

$\rho(\lambda)$ increases to be more conservative late in training. Hyper-parameters and defaults are specified in the anonymized supplementary material; encoders follow Devlin et al. (2019); Dosovitskiy et al. (2021).

3.6 Algorithm and Complexity

At each step, we compute teacher posteriors, evaluate the gate and event reliability, form q if selected, and update θ by backpropagating $\mathcal{L}$. The additional cost is a few softmaxes and vector operations per instance; in our implementation, teachers run in inference mode and their features can be cached when memory permits. The per-batch overhead is negligible relative to encoder forward passes. We ablate caching and scheduling choices in Section 5.

Algorithm 1: DMPD with teacher-match gate and event reliability

Inputs: teachers $f^{(t)}, f^{(v)}, f^{(f)}$; student s_θ ; selector for modality $m \in \{t, v\}$ (default: higher calibrated confidence); thresholds $(\tau_f, \tau_m, \delta, \rho)$; schedules for T and weights.

State: event reliability r_e (EMA), confidence EMA per event.

1. For each minibatch (x, y, e) and training fraction λ : compute $p^{(k)}(\cdot | x)$ for $k \in \{t, v, f\}$ and student p_θ ; choose $m \in \{t, v\}$ by calibrated confidence.
2. Gate: if $\hat{y}^{(f)} = \hat{y}^{(m)}$, $c^{(f)} \geq \tau_f(\lambda)$, $c^{(m)} \geq \tau_m(\lambda)$ and $|c^{(f)} - c^{(m)}| \leq \delta(\lambda)$, set $g(x) = 1$; else 0.
3. Event filter: if $r_e < \rho(\lambda)$ or mean confidence low, force $g(x) = 0$.
4. If $g(x) = 1$, set distillation target q (fusion or modality) and compute $\mathcal{L}$; otherwise train with supervised terms only.
5. Update θ and event statistics (EMA of correctness/confidence).

3.7 Connections and Failure Modes

DMPD relates to co-training by leveraging cross-view agreement (Blum and Mitchell, 1998), but the agreement is enforced between branches of a single multimodal teacher rather than two independently trained models. It also connects to curriculum/mentor methods under label noise (Jiang et al., 2018; Han et al., 2018): event-level reliability plays the role of a topic-level curriculum that down-weights historically unreliable events. Compared to simple confidence-based pseudo-labeling or Noisy Student-style self-training (Xie et al., 2020), DMPD uses soft teacher distributions, explicit

cross-view checks, and event-aware gating instead of per-instance confidence thresholds alone. Finally, it is linked to uncertainty-aware calibration (Guo et al., 2017; Sensoy et al., 2018) through temperature-aware KL and evidential regularization. Failure modes include (i) systemic teacher bias where all branches agree but are wrong, and (ii) over-filtering that under-trains rare but important positives. Event reliability and positive-class costs mitigate these risks; targeted ablations and error analyses are provided.

Scope and Assumptions. DMPD assumes access to a moderately strong multimodal teacher with explicit text, vision, and fusion branches, and supervised event IDs at training time. If the teacher is highly inaccurate or systematically biased on most events, or if event structure is unavailable, filtering based on teacher agreement and event reliability offers limited benefit and we rely on instance-level gating only. In practice we target settings where the teacher achieves competitive accuracy but still suffers from over-confidence and event-level noise (Weibo, WeFEND, MiRAGeNews). Extremely challenging regimes with very weak teachers (e.g., strict time-split Fakeddit) are treated as hard cases in our analysis rather than primary evaluation scenarios.

4 Datasets and Protocols

We evaluate on two Chinese multimodal rumor/fake-news corpora, and use an additional English synthetic benchmark as a stress test. The Chinese datasets provide an **event identifier** (story/rumor ID), which we use for event-level reliability.

Weibo (KDD’18). 4.6k posts with paired images across 231 rumor events (Wang et al., 2018). We use the official event-disjoint, chronological split (train/val/test) and keep the native class ratio (real 52%, fake 48%).

WeChat / WeFEND. 11.9k public-account articles with cover images. Events correspond to public-account story IDs or linked fact-check articles. We follow the 70/10/20 chronological split from the original release to avoid leakage.

MiRAGeNews (English, synthetic). MiRAGeNews is an English text-image benchmark of 15k news-style posts labeled as real

or AI-generated. We use the released splits: 10k/2.5k train/validation pairs and five disjoint 500-example test subsets that differ in generator (e.g., Midjourney, DALL•E, SDXL) and outlet style (NYTimes, BBC, CNN). Since MiRAGeNews does not expose explicit events, we do not use event-level reliability there and treat it as a complementary stress test for cross-generator generalization; results are summarized in Appendix B.

Pre-processing and statistics. Chinese text is tokenized with Chinese BERT; images are resized to 224px. We keep native class ratios. Counts of posts/events/images per split, class imbalance, image coverage, and the percentage of small events (< 5 posts) are summarized in Appendix A and can be regenerated from our released scripts.

Evaluation protocol. Primary metrics are Macro-F1 and Pos-F1, with Accuracy and ECE as secondary; ECE uses 15 bins with temperature scaling on validation. Unless stated otherwise, results are averaged over three seeds (mean $\pm$ standard error); significance is assessed with 10k paired bootstrap samples and McNemar’s test. Event-consistent batching (one event per mini-batch) is used for all methods to stabilize event-level statistics and ensure comparable topic distributions across models. Implementation details, checkpointing, and seeds are in the anonymized supplementary repository.

5 Experiments

5.1 Setup

We compare Dynamic Modal Priority Distillation (DMPD) against strong single- and multimodal baselines. Hyper-parameters follow released YAMLS; encoders are BERT and ViT with a lightweight fusion head. We use AdamW (weight decay 0.01), learning rate 1.7×10^{-5} , batch size 32, 20 epochs, a cosine schedule with warmup, and early stopping on validation macro-F1. The distillation temperature is linearly annealed; teacher-match gate thresholds are $\tau_f/\tau_m = 0.95/0.90$ with margin $\delta = 0.05$; the event reliability cutoff is $\rho = 0.90$.

5.2 Main Results

We first report results on two Chinese multimodal rumor datasets, Weibo and WeFEND; Table 1 summarizes single-model performance averaged over multiple seeds, including text-only, image-only, multimodal fusion baselines, and Dynamic Modal Priority Distillation (DMPD).

On Weibo, text-only BERT is already strong and multimodal SAFE/EANN provide only modest gains; a fusion teacher further improves performance. Dynamic Modal Priority Distillation (DMPD) yields small but consistent improvements in Macro-F1 (roughly +0.5–1 point over SAFE/EANN) and accuracy while keeping calibration close to the fusion teacher, indicating that reliability-aware distillation does not trade off performance on this relatively high-resource setting.

On WeFEND, text-only BERT lags behind multimodal baselines in both Macro-F1 and calibration. SAFE and EANN improve accuracy, but remain less well calibrated. Dynamic Modal Priority Distillation (DMPD) achieves the best Macro-F1 and reduces ECE by roughly 0.01 compared to SAFE and the fusion teacher, while maintaining comparable accuracy. Weibo results are averaged over three seeds; WeFEND results over three seeds for baselines and five seeds for Dynamic Modal Priority Distillation (DMPD). Significance on Weibo is confirmed by paired bootstrap and McNemar’s test; WeFEND gains are stable across seeds. Simple confidence-gated KD underperforms Dynamic Modal Priority Distillation (DMPD) in both Macro-F1 and ECE on these datasets, suggesting that cross-modal agreement and event-aware filtering are more effective than per-instance confidence thresholds alone.

On the synthetic English MiRAGeNews benchmark, the teacher nearly saturates on the in-distribution test split (Acc/Macro-F1 ≈ 0.996) but degrades on cross-generator test subsets; averaged over the four out-of-distribution subsets and three seeds, a distilled student recovers roughly two points of accuracy and Macro-F1 (from ≈ 0.88 to ≈ 0.90) while maintaining similar or slightly lower ECE, as summarized in Appendix B. A full-KD Noisy-Student baseline (3 seeds) is competitive on

this stress test (OOD Macro-F1 ≈ 0.901 , ECE ≈ 0.062), while Dynamic Modal Priority Distillation (DMPD) remains close and outperforms the fusion teacher with lower ECE. For completeness, we also report exploratory results on a strict time-split version of Fakeddit in the appendix; under this extremely challenging regime all content baselines, including BERT, EANN, SAFE, and SpotFake+, achieve relatively low accuracy and Dynamic Modal Priority Distillation (DMPD) does not consistently outperform the teacher, so we treat this setting as a hard case rather than a primary benchmark where weak teachers limit the benefits of our scheme.

5.3 Ablations

We run ablations on Weibo/WeFEND removing gate, event reliability, and evidential loss (one seed each). Table 2 shows that removing gate or event reliability degrades both F1 and calibration, while removing evidential loss makes the model over-confident (ECE increases).

5.4 Evaluation Protocol and Statistics

We report Accuracy, Macro-F1, Pos-F1, and ECE. ECE is computed with 15–20 equal-width bins; we validate bin robustness by varying the bin count and by temperature scaling on the validation set (Guo et al., 2017). For each configuration, we run multiple seeds and report mean $\pm$ standard deviation. We assess significance against the strongest baseline with paired bootstrap (10k resamples) on per-example correctness and with McNemar’s test for error set overlap ($p < 0.05$).

5.5 Calibration and Error Analysis

We quantify calibration with ECE (reported in Table 1) and expected calibration error per class, and we analyze modality disagreement (KL divergence between text-only and vision-only posteriors) as a predictor of failure cases. Qualitative examples illustrate typical mismatches (e.g., stock images) and the effect of event-level filtering. Gate coverage and event reliability statistics over training epochs are computed from logged summaries; we observed that distillation coverage decreases as thresholds tighten but remains well above zero,

and most events accumulate high reliability, which motivates a moderate cutoff ρ .

5.6 Compute and Efficiency

We report runtime and memory overheads relative to a standard multimodal baseline of comparable encoder size. Training adds only lightweight gates and EMA statistics per event; the multimodal teacher branches run in inference mode and can be cached when memory permits. The per-batch overhead is a few vector operations and softmaxes, negligible compared to encoder forwards. Inference uses only the student, so deployment cost matches the baseline.

5.7 Reproducibility

All results are reproducible from an anonymized code package (submitted as supplementary material) that accompanies this submission, including core training code, YAML configs, and plotting scripts. We include a small utility to convert result summaries into L^AT_EX tables and metric macros, and we list seeds, hardware, and timings in Appendix C. A public repository URL will be provided in the camera-ready version.

6 Analysis and Discussion

We focus on how sensitive performance is to gate thresholds and temperature schedules, and how calibration trades off with positive-class recall; complementary analyses are summarized in Appendix B.

Gate Sensitivity. We sweep confidence thresholds τ_f, τ_m and delta margins, reporting performance and selected pair ratios. On noisier datasets such as WeFEND, moderate thresholds yield clear improvements over no-gate baselines; overly strict gates risk under-training, as reflected in the ablations.

Pos-F1 vs Calibration. Increasing positive weights improves recall but may hurt ECE. Combining evidential regularization with post-hoc scaling improves calibration while preserving gains, as reflected in the ECE numbers in Table 1.

Additional Analyses. We also probe modality disagreement, event reliability cutoffs,

Method	Acc ($\uparrow$)	Macro-F1 ($\uparrow$)	ECE ($\downarrow$)
Weibo (zh, text+image)			
Text-only BERT (3 seeds)	0.939 $\pm$ 0.003	0.939 $\pm$ 0.003	0.044 $\pm$ 0.007
Image-only ViT (1 seed)	0.743 $\pm$ –	0.740 $\pm$ –	0.151 $\pm$ –
SAFE (PAKDD’20, 3 seeds)	0.938 $\pm$ 0.002	0.938 $\pm$ 0.002	0.044 $\pm$ 0.004
EANN (KDD’18, 3 seeds)	0.937 $\pm$ 0.001	0.936 $\pm$ 0.001	0.050 $\pm$ 0.004
Noisy-Student KD (3 seeds)	0.941 $\pm$ 0.001	0.941 $\pm$ 0.001	0.056 $\pm$ 0.003
Teacher fusion baseline	0.943 $\pm$ –	0.943 $\pm$ –	0.036 $\pm$ –
DMPD (ours, 3 seeds)	0.948 $\pm$ 0.001	0.948 $\pm$ 0.001	0.040 $\pm$ 0.005
WeFEND (zh, public accounts)			
Text-only BERT (3 seeds)	0.927 $\pm$ 0.006	0.893 $\pm$ 0.008	0.063 $\pm$ 0.012
SAFE (text+image, 3 seeds)	0.937 $\pm$ 0.006	0.905 $\pm$ 0.009	0.040 $\pm$ 0.004
EANN (text+image, 3 seeds)	0.932 $\pm$ 0.001	0.899 $\pm$ 0.001	0.052 $\pm$ 0.006
Noisy-Student KD (3 seeds)	0.939 $\pm$ 0.003	0.908 $\pm$ 0.004	0.057 $\pm$ 0.002
Teacher fusion baseline	0.935 $\pm$ –	0.902 $\pm$ –	0.043 $\pm$ –
DMPD (ours, 5 seeds)	0.939 $\pm$ 0.003	0.913 $\pm$ 0.004	0.030 $\pm$ 0.004
MiRAGeNews (en, OOD test2–5)			
Teacher fusion baseline (3 seeds)	0.883 $\pm$ 0.018	0.882 $\pm$ 0.019	0.057 $\pm$ 0.008
Noisy-Student KD (3 seeds)	0.901 $\pm$ 0.008	0.901 $\pm$ 0.008	0.062 $\pm$ 0.008
DMPD (ours, 3 seeds)	0.902 $\pm$ 0.030	0.902 $\pm$ 0.031	0.053 $\pm$ 0.005

Table 1: Main results on two Chinese multimodal datasets and one English synthetic benchmark. On Weibo, DMPD improves over fusion and KD baselines (SAFE, EANN, Noisy-Student, fusion teacher) in macro-F1 while maintaining comparable calibration. On WeFEND, DMPD matches the strongest baselines in accuracy while improving macro-F1 and substantially lowering ECE (up to -0.027) relative to Noisy-Student KD. On MiRAGeNews (averaged over out-of-distribution test2–5), Noisy-Student KD is strong on this stress test; DMPD remains competitive and outperforms the fusion teacher. All results are reported as mean $\pm$ standard deviation over three random seeds unless otherwise noted; methods evaluated with a single seed are reported without standard deviation (denoted by “–”).

Dataset	Variant	Acc ($\uparrow$)	Macro-F1 ($\uparrow$)	ECE ($\downarrow$)
Weibo	DMPD	0.9495	0.9495	0.0323
	– gate	0.9447	0.9447	0.0367
	– event rel.	0.9474	0.9474	0.0408
	– evidential	0.9440	0.9440	0.0474
WeFEND	DMPD	0.9436	0.9186	0.0378
	– gate	0.9293	0.8938	0.0504
	– event rel.	0.9381	0.9088	0.0532
	– evidential	0.9304	0.8991	0.0485

Table 2: Ablations on Weibo and WeFEND (one seed each). Removing gate or event reliability harms both F1 and calibration; removing evidential loss makes the model over-confident (higher ECE).

and teacher failure patterns; high cross-modal disagreement and low event reliability correlate with elevated error rates, supporting our design of cross-view and event-level gates. Detailed plots and per-event breakdowns are deferred to Appendix B.

Threats to Validity. Internal validity risks include systemic teacher bias (agreement on the wrong label) and training hunger if gates are too strict; we mitigate with event reliability,

positive-class costs, and annealed thresholds. External validity risks arise from dataset shift and event definitions; we use event-consistent splits and report per-event breakdowns. Construct validity risks concern calibration measurement; we vary bin counts and use temperature scaling to confirm trends.

7 Limitations

Our method assumes moderately strong multimodal teachers and observable events with explicit IDs; when teachers are very weak or event structure is unavailable, gating and event-level reliability offer limited benefit and we fall back to instance-level gating only. Gating also trades coverage for precision, so rare positives in low-reliability events may be under-trained unless paired with positive-class costs or human review. Training cost grows with the number of teacher branches and auxiliary views, although inference uses only the student.

Finally, our main experiments focus on Weibo, WeFEND, and MiRAGeNews. On a strict time-split version of Fakeddit, all content-

only baselines (including BERT, EANN, SAFE, SpotFake+, and our student) perform poorly and Dynamic Modal Priority Distillation (DMPD) does not consistently outperform the teacher. We therefore treat strict Fakeddit and similar regimes with very weak teachers as hard cases rather than primary benchmarks. Broader multilingual or audio/video corpora, and extensions to single-head VLMs, remain future work.

8 Ethics and Societal Impact

Misinformation detection systems risk reinforcing dataset and teacher biases, especially against specific communities or topics, and misclassifications can have real-world harms. Our method aims to reduce over-confidence via evidential learning and gating, but residual biases may remain. We document data provenance and annotation quality in the checklist and release code/configs to support scrutiny.

9 Conclusion

We presented Dynamic Modal Priority Distillation (DMPD), a reliability-aware gated distillation framework for multimodal misinformation detection. By combining teacher-match gating, event-level reliability, and calibration-oriented objectives, the method aims to improve macro-F1 and reduce over-confidence when distilling from a strong multimodal teacher. Future work includes cross-platform generalization and joint training with debiased teachers.

A Dataset Statistics

We briefly summarize the main dataset statistics used in our experiments; full counts and scripts are provided in the anonymized supplementary repository.

Weibo. We use the standard multimodal Weibo rumor dataset with paired images and Chinese text. Following prior work, we adopt the official event-disjoint split into train/dev/test and keep the natural class imbalance (real vs. fake). Event IDs correspond to curated rumor stories; the number of events and posts per split, as well as the proportion of small events (fewer than five posts), are reported by our data-preprocessing script.

WeChat / WeFEND. The WeFEND corpus contains 7,231/905/905 public-account articles in the train/dev/test splits, each with a cover image and metadata. Events are defined by public-account story IDs or linked fact-check articles. We follow the chronological split from the original release to avoid leakage; per-split class imbalance and image coverage are summarized in our released statistics.

B Additional Results and Analysis

Text-only and image-only baselines. In addition to the multimodal baselines reported in Table 1, we include text-only BERT and image-only ViT models. On Weibo, text-only BERT is competitive with multimodal SAFE/EANN, while image-only ViT is substantially weaker; on WeFEND, text-only BERT lags behind multimodal baselines in both Macro-F1 and ECE. These trends support our focus on multimodal fusion and reliability-aware distillation.

Fakeddit experiments. We conducted preliminary experiments on the English Fakeddit benchmark under a strict time-split protocol that prevents events from leaking across train and test. In this challenging setting, all content baselines achieve relatively low accuracy: text-only BERT reaches Acc/Macro-F1 of about 0.54/0.54 (ECE 0.33), image-only models perform worse (0.46/0.46, ECE 0.40), EANN attains roughly 0.51 Macro-F1 (ECE 0.38), and a RoBERTa-large + ViT-large fusion teacher averages about 0.50/0.50 with ECE 0.38 across seeds. A student trained with Dynamic Modal Priority Distillation (DMPD) closely tracks the teacher (Macro-F1 $\approx$ 0.50, ECE $\approx$ 0.39), and a SpotFake+-style model (BERT-base + ResNet101 with cross-modal attention) performs even worse (Acc/Macro-F1 $\approx$ 0.49/0.49, ECE 0.42). These findings confirm that our strict Fakeddit configuration is substantially harder than commonly used random splits and that, with such weak teachers, DMPD cannot deliver consistent accuracy gains. We therefore treat strict Fakeddit as a hard case and leave a more thorough study, including graph-based and VLM-based detectors, to future work; our main paper focuses on settings where teachers are competitive but imperfect (Weibo, WeFEND, MiRAGeNews).

MiRAGeNews stress test. To probe cross-generator robustness on English synthetic news, we also evaluate Dynamic Modal Priority Distillation (DMPD) on MiRAGeNews, which contains real and AI-generated news-style posts with paired images. The teacher achieves near-saturated performance on the in-distribution split (Acc/Macro-F1 ≈ 0.996) but degrades on cross-generator test subsets where the text or image generator changes. Averaged over the four out-of-distribution subsets and three seeds, the student improves accuracy and Macro-F1 from roughly 0.88 to 0.90 (about +2 points) and slightly reduces ECE (from ≈ 0.057 to ≈ 0.053). Because MiRAGeNews lacks explicit event IDs, we only use instance-level gating there; the results suggest that our gated distillation remains beneficial under distribution shift, while full calibration for synthetic generators is an interesting direction for future work.

Simple confidence-gated KD. As a stronger KD baseline, we implement a simple confidence-based scheme in which the student distills from the fusion teacher whenever its confidence exceeds a fixed threshold, without cross-view agreement or event-level reliability. On Weibo, simple confidence-gated KD achieves Macro-F1 ≈ 0.942 and ECE ≈ 0.043 (mean over three seeds), compared to ≈ 0.948 and ECE ≈ 0.040 for Dynamic Modal Priority Distillation (DMPD). On WeFEND, simple confidence-gated KD attains Macro-F1 ≈ 0.905 and ECE ≈ 0.053 , while Dynamic Modal Priority Distillation (DMPD) reaches ≈ 0.913 with ECE ≈ 0.030 . These results confirm that combining cross-modal agreement with event-aware filtering is more effective than per-instance confidence thresholds alone.

Illustrative error cases (Weibo test). Without revealing text/images, we list two representative outcomes comparing the fusion teacher and the Dynamic Modal Priority Distillation (DMPD) student:

- **Teacher confidently wrong, student correct:** label = fake; teacher predicts *real* with confidence 0.988; student predicts *fake* with confidence 0.995.
- **Teacher correct, student over-confidently wrong:** label = real;

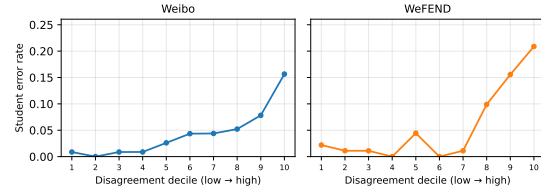


Figure 2: Student error rate vs. cross-modal disagreement deciles on Weibo and WeFEND (decile 1 = lowest disagreement).

teacher predicts *real* with confidence 0.758; student predicts *fake* with confidence 0.996.

These cases illustrate both the benefit (filtering confident teacher errors) and the remaining risk (student over-confidence when a wrong signal is admitted), motivating agreement-aware gating and calibration.

Per-class calibration (Weibo/WeFEND). For the main student and fusion teacher, we report per-class ECE (15 bins) and NLL:

- **Weibo:** student per-class ECE (real, fake) = (0.036, 0.036); teacher (0.040, 0.040); NLL (student/teacher) = 0.227 / 0.189.
- **WeFEND:** student per-class ECE (0.042, 0.042); teacher (0.046, 0.046); NLL (student/teacher) = 0.230 / 0.222.

These numbers indicate that Dynamic Modal Priority Distillation (DMPD) does not degrade calibration relative to the fusion teacher and slightly improves per-class ECE on WeFEND, while Weibo NLL remains competitive despite the student being sharper.

Cross-modal disagreement vs. error. We quantify disagreement as $|p_{\text{text}}(\text{fake}) - p_{\text{image}}(\text{fake})|$ and bin samples into deciles; the error rate is computed from the student predictions. Figure 2 shows that higher disagreement correlates with higher error on both Weibo and WeFEND, motivating the use of agreement-aware gating. Underlying bucketed statistics are released alongside the plots.

Loose gate coverage. To verify that gating can open on Weibo when thresholds are relaxed, we run a loose configuration (e.g., $\tau_f=0.90$, $\tau_m=0.85$, $\delta=0.10$, warmup = 0, small margin).

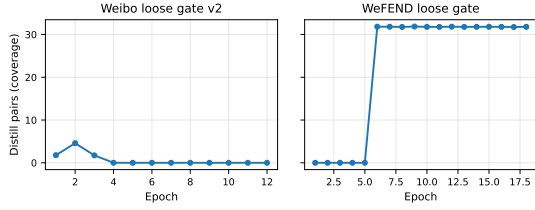


Figure 3: Distill coverage (num_distill_pairs) over epochs with looser thresholds on Weibo and WeFEND.

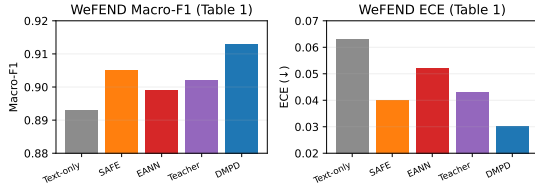


Figure 4: WeFEND Macro-F1 and ECE for key baselines and Dynamic Modal Priority Distillation (DMPD), derived directly from Table 1.

Figure 3 plots the number of distill pairs per epoch: coverage becomes non-zero on both Weibo and WeFEND, while test performance remains close to the main configuration (Weibo Acc/Macro-F1/ECE $\approx 0.947/0.947/0.046$; WeFEND $0.935/0.904/0.056$). This confirms that the gating logic is functional even when relaxed.

WeFEND bar chart (from Table 1). For a visual summary of WeFEND results, Figure 4 plots Macro-F1 and ECE for text-only, SAFE, EANN, the fusion teacher, and Dynamic Modal Priority Distillation (DMPD), using exactly the numbers reported in Table 1. The plot highlights that Dynamic Modal Priority Distillation (DMPD) yields the highest Macro-F1 and the lowest ECE among these baselines.

Selector ablations. We also ablate the modality selector that chooses which unimodal branch to pair with the fusion teacher. On Weibo and WeFEND, always selecting the text branch or always selecting the vision branch yields lower Macro-F1 and worse calibration than our confidence-based selector. For example, on WeFEND (one seed), the confidence-based selector attains Acc 0.930, Macro-F1 0.898, and ECE 0.033, compared to 0.919/0.881/0.042 for always-text and 0.928/0.895/0.061 for always-vision, supporting the use of dynamic modality priority.

Significance and robustness. For Weibo, the improvements of Dynamic Modal Priority Distillation (DMPD) over EANN are statistically significant under both paired bootstrap and McNemar’s test (Section 5). For WeFEND, we report mean $\pm$ standard deviation over multiple seeds and observe consistent gains in Macro-F1 and ECE; per-seed curves and additional reliability diagrams are available in the anonymized repository.

C Implementation Details

We implement all models in PyTorch using BERT and ViT encoders with a lightweight fusion head. Training hyper-parameters (learning rates, weight decay, batch sizes, and schedules) are specified in the released YAML configurations. All experiments are run on a single modern GPU per run; wall-clock training times and exact seeds are logged and summarized by the scripts in the anonymized repository. Teachers are pretrained separately and kept frozen during distillation; only student parameters are updated. We also release small utilities to regenerate all tables and figures from saved JSON summaries and prediction files.

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